

INDIAN INSTITUTE OF TECHNOLOGY DELHI

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INTERNSHIP REPORT

Arduino Based Embedded Systems

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Contents

List of Figures

List of Tables

Chapter 1

Introduction

This internship focused on the design and implementation of embedded systems using the Arduino UNO platform. The experiments cover basic digital interfacing, sensor integration, display modules, automation, and embedded applications.

The objective of the internship was to strengthen practical knowledge of microcontrollers, embedded programming, hardware interfacing, and real-time control systems through hands-on projects.

The report documents each experiment with its objective, circuit design, implementation methodology, working principle, results, and conclusion.

Chapter 2

Basic Arduino Experiments

This chapter presents the fundamental Arduino-based experiments carried out during the internship. These experiments introduce basic digital interfacing, display modules, and embedded programming concepts using the Arduino UNO development board. The knowledge gained from these experiments forms the foundation for more advanced embedded system applications discussed in later chapters.

2.1 LED Flash System

2.1.1 Objective

The objective of this experiment is to interface an LED with the Arduino UNO and make it blink continuously at a fixed time interval using digital output pins.

2.1.2 Problem Statement

Design and implement a simple LED flashing system using Arduino UNO to understand digital output control, GPIO programming, and timing functions.

2.1.3 Components Used

2.1.4 Hardware Connections

2.1.5 Circuit Diagram

2.1.6 Software Used

- Arduino IDE
- SimulIDE

2.1.7 Source Code

The complete Arduino source code for this experiment is available in the project repository.

[Click here to view the Arduino source code](#)

2.1.8 Working Principle

The Arduino UNO configures digital pin D12 as an output during initialization. The program repeatedly changes the output state of the pin from HIGH to LOW with a one-second delay between each transition. As a result, the connected LED turns ON and OFF continuously, demonstrating the basic operation of digital output control and timing functions in Arduino programming.

2.1.9 Results

The LED blinked successfully with a one-second ON period followed by a one-second OFF period. The experiment verified the correct operation of Arduino digital output pins and demonstrated the use of software-based timing control.

2.1.10 Conclusion

This experiment successfully demonstrated the interfacing of an LED with Arduino UNO. It provided practical understanding of GPIO configuration, digital output control, and program execution flow, forming the basis for subsequent embedded system experiments.